OLYMPIC SPORTS ANALYSIS

Data Analyst Project — Final Report

A professional, source-grounded review of Olympic Games history, sport and event structure, athlete participation, and medal outcomes.

PROJECT EVIDENCE

SQL / Excel outputs • Power BI solved questions • Complete dashboard image appendix

Prepared: October 2026

REPORT GUIDE

Contents and reading guide

The report is split into an analytical narrative followed by a visual appendix. All pages from the supplied Power BI dashboard PDF are reproduced in the appendix; all 18 solved-question visual pages are also retained for traceability.

Section	Purpose
1. Executive summary	Headline findings and baseline KPIs.
2. Project brief and objectives	Business question, scope, and intended audience.
3. Data and methodology	Inputs, workflow, aggregation approach, and grain cautions.
4. Games architecture	Summer/Winter count and hosting context.
5. Sports and events	Reported event-count distribution and event categories.
6. Athlete participation	Gender participation, age context, and participation interpretation.
7. Medal outcomes	Gold and total medal leaderboard interpretation.
8. Participation by country	Reported participation ranking and how to use it.
9. Power BI dashboard review	Dashboard page roles and improvement suggestions.
10. Recommendations	Prioritized work to improve reliability and usability.
11. Limitations and metric governance	Definitions and inconsistencies to validate.
12. Conclusion	Final assessment and next steps.
Appendix A	All supplied dashboard pages, reproduced as full-page images.
Appendix B	All 18 Power BI solved-question pages, reproduced as full-page images.

How to interpret figures: The report uses the terminology and values present in the supplied project files. Rounded summary figures and visual-level figures may have different aggregation rules. Where the files do not define a metric precisely, the report flags it for validation instead of silently changing it.

01 / EXECUTIVE SUMMARY

Key findings at a glance

The supplied SQL/Excel and Power BI materials support a descriptive historical review of the Olympic Games. The findings are useful for exploration and portfolio presentation, subject to metric-definition checks.

51	66	757	129K
Games in baseline	Sports in MECE baseline	Events in MECE baseline	Athlete participations

Finding	Evidence from supplied outputs	Analyst interpretation
Games by season	29 Summer; 22 Winter	Summer represents 56.9% of the 51 Games; Winter 43.1%.
Event categories	458 men's; 211 women's; 88 mixed/open	The historical aggregate is male-event weighted; not a current-edition comparison.
Participation by gender	95K male (73.9%); 34K female (26.1%)	A historical aggregate across eras; time-based breakdown is needed to study change.
Top reported gold-medal NOC	USA: 2,533; Soviet Union: 1,135; Germany: 843	Raw totals depend on exposure, event portfolio, team-event rules, and NOC history.
Top reported participation NOC	USA: 9,225; UK: 5,783; France: 5,198	Participation volume and medal outcomes should be analysed as separate measures.

Executive conclusion. The project has a useful descriptive BI structure. Its next maturity step is metric governance: define each count's grain, reconcile source differences, and ensure every headline KPI can be traced to a validated source query.

02 / PROJECT BRIEF

Business question, scope, and objectives

Primary question: How have the Olympic Games changed over time, and what patterns are visible in sport/event composition, athlete participation, and medal outcomes?

The project is framed as a descriptive analytics exercise that combines SQL/Excel question outputs with a Power BI exploration layer. It is intended for a data analyst portfolio, interview walkthrough, and stakeholder-style review—not as a causal or predictive study.

Objective area	Questions addressed
Games architecture	How many Games are represented by season? What host/temporal patterns appear in the supplied outputs?
Sport/event portfolio	Which sports have the highest reported event counts? How are events distributed across gender categories?
Participation	How are athlete participations distributed by gender, age, country, and year?
Medal outcomes	Which NOCs lead the reported gold and total medal rankings? How should those rankings be interpreted?
Decision support	How can dashboard slicers help users narrow and compare the data?

The source MECE report organizes the analysis into four pillars: Games architecture and temporal evolution; sports and event portfolio dynamics; demographic profile and participation trends; and geopolitical dominance and medal outcomes. This report preserves those pillars.

03 / DATA INVENTORY

Source files and their analytical role

Input	What it contains	How it is used
SQL/Excel workbook	18 answer tabs with query/grouped outputs.	Year/season summaries, sport counts, participation outputs, and medal comparisons.
Power BI solved-questions PDF	18 question-specific visual pages.	Traceable visuals for sports, events, gender, participation, and medal rankings.
Power BI dashboard PDF	5 dashboard pages.	Evidence of the interactive dashboard structure and its visual design.
MECE breakdown report	Four analytical pillars and baseline KPIs.	Baseline values: 51 Games, 66 sports, 757 events, 129K participations, 228 NOCs, 39K medals.

The report does not introduce outside data. All figures below are transcribed from the supplied summaries or visual outputs. Exact row-level validation should be performed against the underlying dataset before publishing the numbers as official statistics.

04 / METHODOLOGY

Analytical workflow and counting grain

Step	Analyst activity	Quality control
1. Inspect	Review workbook answer tabs and supplied Power BI visual pages.	Keep source terminology; distinguish workbook results from dashboard summary values.
2. Structure	Group questions into the four MECE pillars.	Avoid repeating the same KPI as separate findings without explaining the difference.
3. Summarize	Use descriptive counts, sums, averages, and ranked comparisons.	Do not claim statistical significance or causation where none is shown.
4. Visualize	Use charts for season counts, event mix, top sports, and country rankings.	Label units and distinguish event counts, participations, medalists, and medals.
5. Interpret	Translate outputs into findings and recommendations.	Flag metric differences and historical NOC naming issues.

Grain matters: a row may represent an athlete-event participation, a medal, an event, a Games edition, or a country-level aggregate. The meaning of a count depends on that grain. A person can appear in multiple editions or events, so participation totals should not automatically be called unique athletes.

05 / PILLAR I — GAMES ARCHITECTURE

Season balance and historical footprint

The MECE baseline lists 51 Games: 29 Summer and 22 Winter. This is a count of editions, not a measure of audience, cost, athlete volume, or international reach.

Season	Games	Share of baseline
Summer	29	56.9%
Winter	22	43.1%
Total	51	100.0%

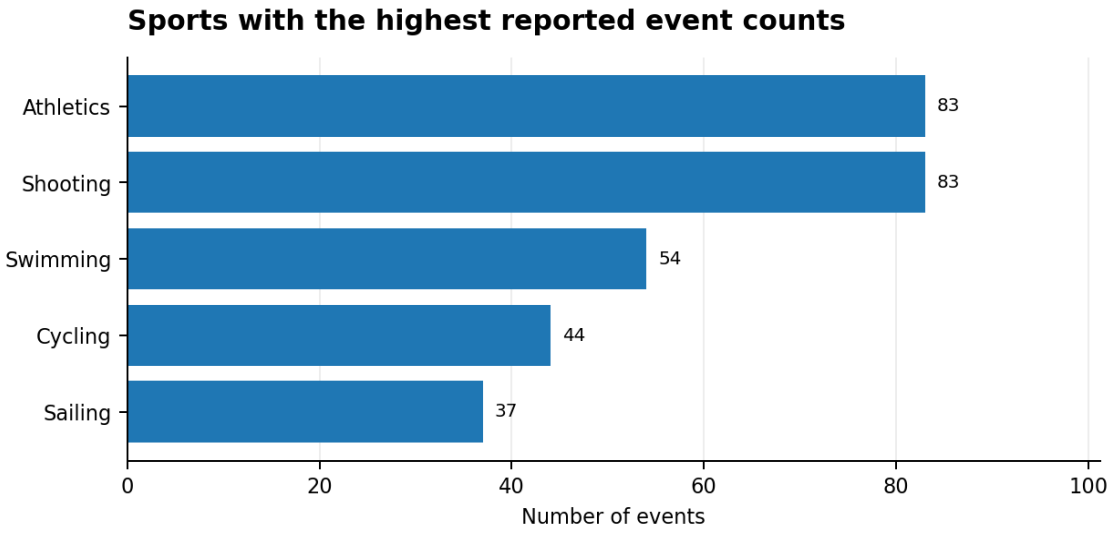
The supplied solved-question visuals show repeated hosts, including Athina and London with three appearances in the MECE summary, and multiple cities with two appearances. Host recurrence is useful context for infrastructure and legacy discussions, but the source materials do not establish that hosting causes better sporting outcomes.

Recommended next step: compare participation, event counts, and NOC representation by Games year, annotate exceptional/cancelled editions where applicable, and standardize host-city naming before using host frequency as a KPI.

06 / PILLAR II — SPORTS PORTFOLIO

Reported sports with the highest event counts

The supplied solved-question output reports Athletics and Shooting with 83 events each, followed by Swimming (54), Cycling (44), and Sailing (37). These are event-count values as displayed in the source visual, not counts of athlete starts, medals, or annual events.

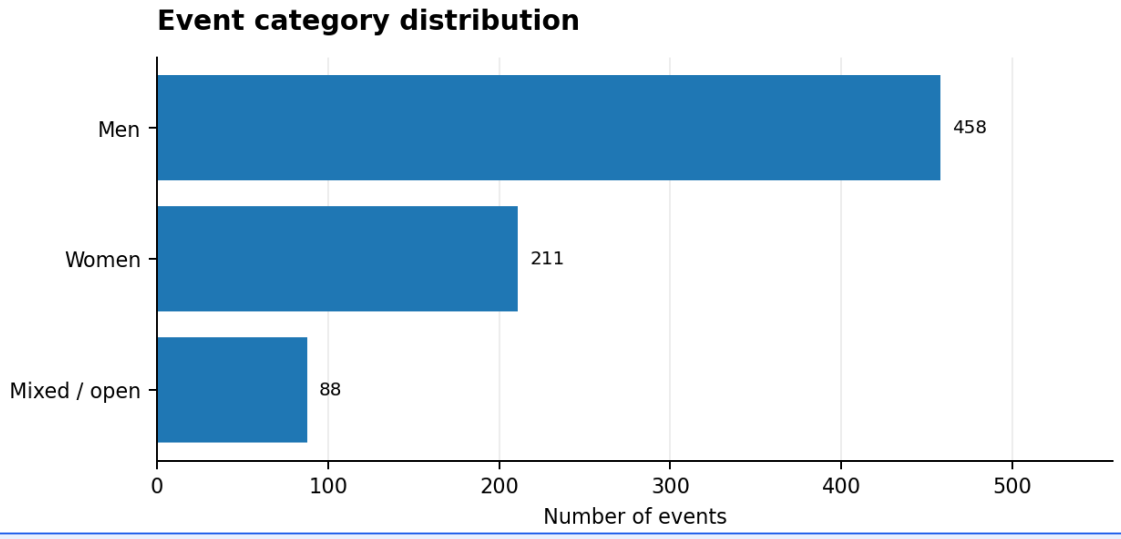


The concentration of events in several sports suggests that a single total-medal ranking can be influenced by the size of each sport’s event portfolio. A fairer comparison of sport-level medal productivity would require a clearly defined denominator such as events contested or athlete entries.

07 / PILLAR II — EVENT MIX

Distribution across gender categories

The supplied solved-question visual reports 458 men’s events (60.5%), 211 women’s events (27.9%), and 88 mixed/open events (11.6%). These figures describe the accumulated historical dataset, not the gender balance of a single recent edition.

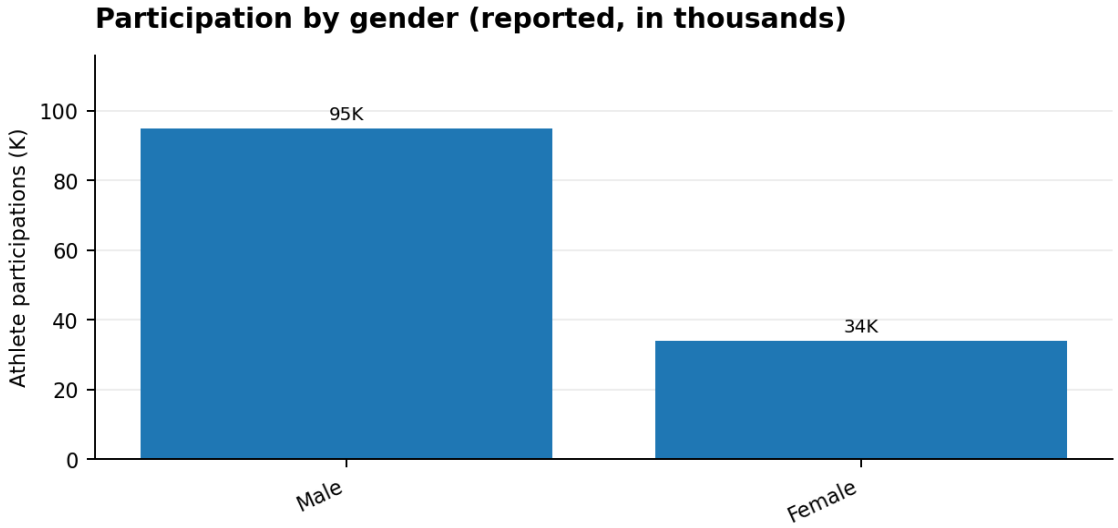


A stronger trend analysis would break these event counts down by Games year, season, and sport. It should show absolute counts alongside percentages so that changes in the overall event programme do not distort interpretation.

08 / PILLAR III — PARTICIPATION PROFILE

Gender participation and historical context

The supplied summary reports 95K male participations (73.9%) and 34K female participations (26.1%). The MECE baseline reports 129K athlete participations overall. Rounded figures may not sum exactly to the displayed total.



Because this is an all-time aggregate, it combines eras with different eligibility rules and event structures. It is useful as a historical baseline, but should not be presented as a current-Games snapshot.

The MECE summary reports a mean athlete age of 25.78 years and identifies ages 18–24 as the largest cohort in its age grouping. A deeper view should examine age by sport and season, because athlete age profiles can differ materially between sports.

09 / PILLAR III — PARTICIPATION BY COUNTRY

Where reported participation is concentrated

The supplied solved-question visual reports the following top country/NOC participation totals. These values should be read as participations as defined by the source output, not necessarily unique people.

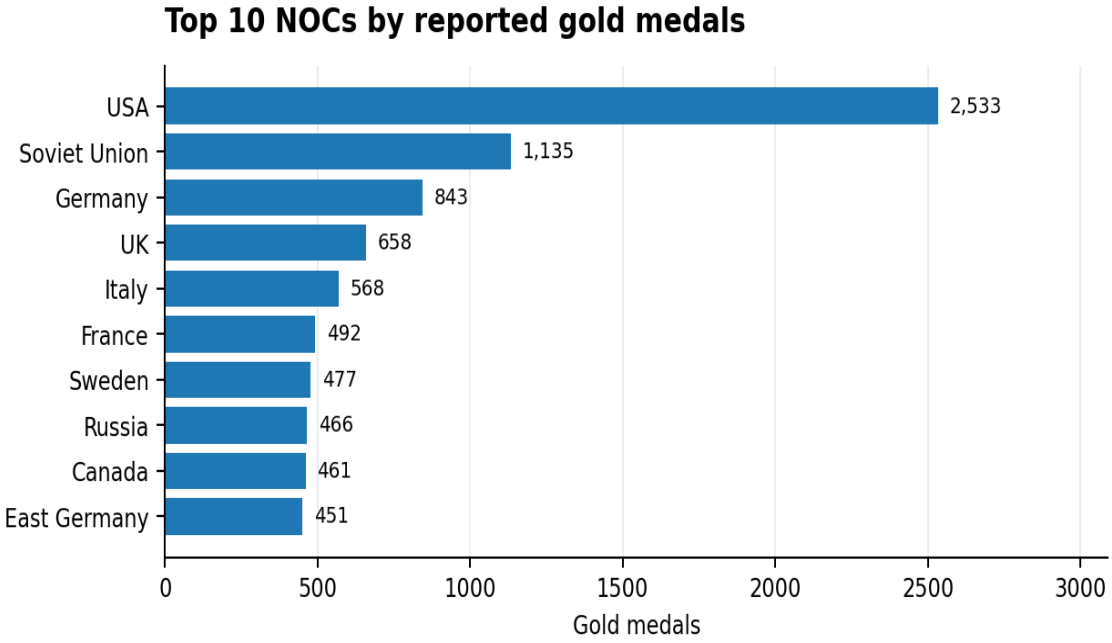
Rank	Country / NOC	Reported participations
1	USA	9,225
2	UK	5,783
3	France	5,198
4	Germany	4,763
5	Canada	4,657
6	Italy	4,654
7	Japan	3,965
8	Sweden	3,781
9	Australia	3,747
10	Poland	2,957

The close reported totals for Canada and Italy show why a report should avoid overemphasizing small rank differences. A country-level dashboard should make it easy to compare participation and medals separately and filter both by Games year.

10 / PILLAR IV — MEDAL OUTCOMES

Gold medal leaderboard

The source output places the USA first in reported gold medals (2,533), followed by the Soviet Union (1,135) and Germany (843). The leaderboard also includes UK, Italy, France, Sweden, Russia, Canada, and East Germany in the top ten.



Raw medal totals are affected by the number of Games attended, event availability, team-event counting conventions, and historical NOC transitions. They are descriptive totals, not normalized measures of national performance.

11 / MEDAL ANALYSIS

Why raw medal counts need context

The source reports 5,414 total medals for the USA, 2,658 for the Soviet Union, and 2,395 for Germany. Medal counts can be reported at different grains—individual medal records, event medals, or team-event medals—so the exact rule must be documented.

Comparison option	What it can answer	Caution
Raw medals	Which NOCs accumulated the most medals in the source dataset?	Favors longer participation history and larger event exposure.
Gold medals	Which NOCs have the most first-place finishes under the source rules?	Gold totals are not equivalent to total medals or overall participation.
Medals per participation	How many medals are associated with each participation under a chosen grain?	Requires compatible numerator/denominator and consistent handling of team events.
Medals by Games edition	How did medal outcomes vary over time?	Should be filtered by season and edition; historical NOC names need care.

Recommended approach: show raw totals for transparency, then provide one normalized measure only after documenting its formula and validating the row-level logic.

12 / BI PRODUCT REVIEW

Power BI dashboard structure

Page	Purpose	Suggested improvement
Olympic overview	Baseline KPIs, Games and host context.	Add clear KPI definitions and a dynamic subtitle showing selected filters.
Sports & events	Explore sports and event counts.	Clarify whether counts refer to events in the full history or the selected Games.
Participant demographics	Explore gender, age, and participation patterns.	Show year-level trend and absolute counts beside percentages.
Medal analysis	Compare medals across NOCs and sports.	Clarify gold vs total medal measures and team-event rules.
Regional analysis	Compare participation and medal outcomes by NOC/region.	Document historical NOC mapping and provide drill-through.

The five-page dashboard structure aligns well with the four analytical pillars. Slicers for Gender, Sport Name, Season, and Games Year support useful exploration. A production-quality dashboard should also expose source/refresh date, metric definitions, and a visible note on historical NOC treatment.

13 / RECOMMENDATIONS

Prioritized next steps

Priority	Action	Value
P1	Create a data dictionary defining Games, events, athlete participations, unique athletes, medalists, and medals.	Prevents misleading comparisons and supports auditability.
P1	Validate headline KPIs against the underlying dataset and query outputs.	Ensures summary cards reconcile with detail visuals.
P1	Standardize historical NOC identities and document successor-state treatment.	Makes country rankings reproducible.
P2	Add gender/event/participation trends by Games year and sport.	Shows how representation changes over time.
P2	Add carefully defined normalized medal metrics alongside raw totals.	Adds context without hiding raw counts.
P2	Improve dashboard tooltips, source notes, and filter-state summaries.	Makes the dashboard easier to interpret and present.
P3	Consider advanced trend or predictive work only after metric validation.	Avoids implying causal or predictive capability without evidence.

14 / LIMITATIONS AND METRIC GOVERNANCE

What should be validated before publication

Issue	Why it matters	Treatment
Sports totals differ	MECE baseline reports 66 sports; one visual shows 52 Summer + 17 Winter sport counts (69 season-specific counts).	Confirm overlap or definition differences; do not force totals to match.
Participation vs unique athletes	A person can appear across multiple Games/events.	Use “participations” unless unique person IDs are consistently deduplicated.
Medal grain	Team events and event-level vs athlete-level medal rows can change totals.	Document medal counting logic.
Historical NOC identities	Soviet Union, East/West Germany, Russia and other names vary by period.	State the mapping policy used for comparison.
Rounded baselines	129K participations and 39K medals are rounded summary figures.	Use rounded values for executive display; validate exact values before publishing.
Descriptive analysis only	Supplied materials show descriptive query outputs and visuals.	Do not claim causal findings or validated forecast performance.

Confidence statement: the directional patterns are supported by the supplied summaries. Exact comparisons across time and NOCs should remain provisional until metric grain, source definitions, and historical identity rules are validated.

15 / CONCLUSION

Final project assessment

The project combines SQL/Excel question outputs with a Power BI dashboard to tell a coherent story about Olympic history. Its strongest contribution is a structured descriptive view of Games, sport/event composition, participation, and medal outcomes.

The key analytical lesson is that participation and medal outcomes must be interpreted in the context of a changing event portfolio, historical eligibility patterns, NOC identity, and counting rules. Clear visuals are important, but clear metric definitions are what make the findings trustworthy.

Project component	Supplied evidence
SQL/Excel analysis	18 answer tabs in the workbook.
Power BI solved questions	18 visual pages supplied.
Power BI dashboard	5 original dashboard pages supplied and reproduced in Appendix A.
Final narrative	This report includes findings, interpretation, recommendations, and limitations.

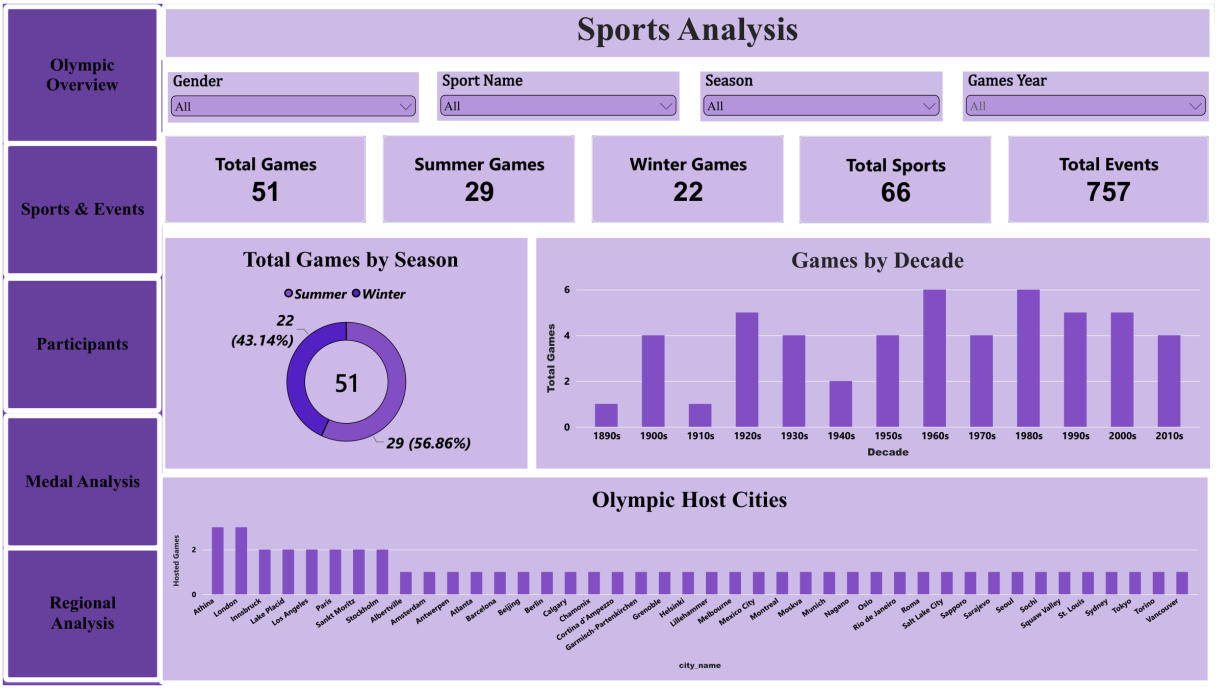
Next best step: validate the data dictionary and reconcile the few source-level count differences before treating the report as a final external/publication-ready analysis.

APPENDIX A

Original Power BI dashboard — all pages

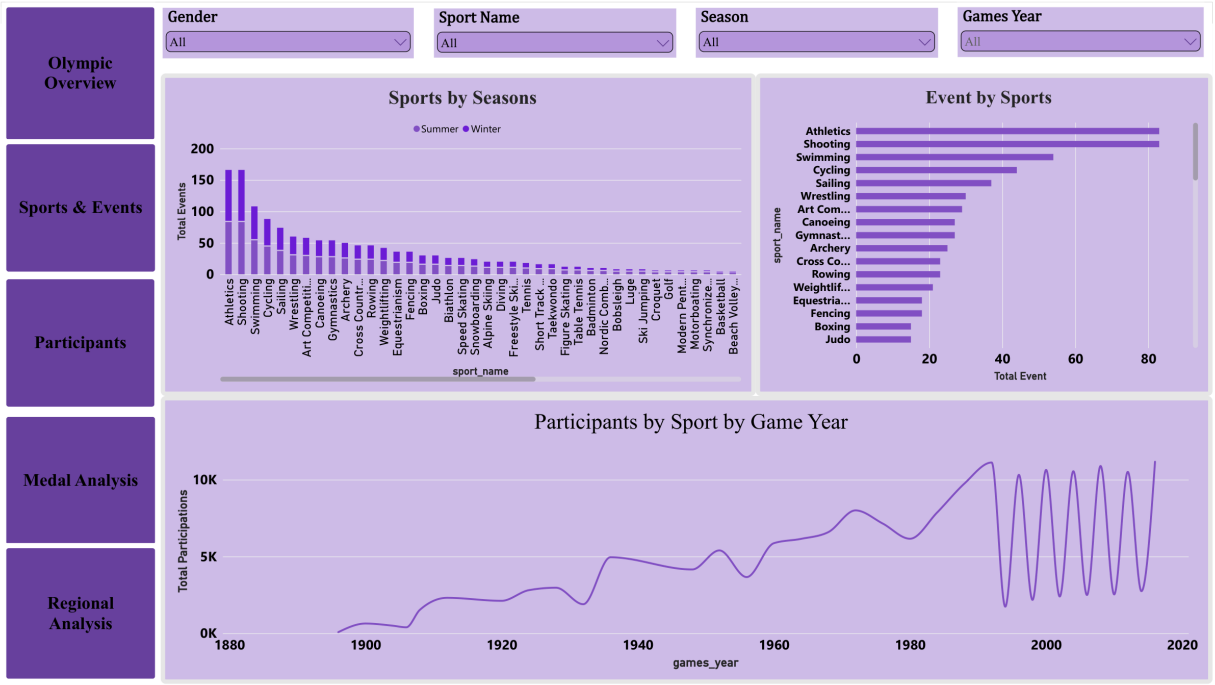
The following pages reproduce every page from the supplied Power BI dashboard PDF. Images are rendered directly from the source PDF to preserve the original chart content, layout, and colors.

Dashboard source page 1 of 5



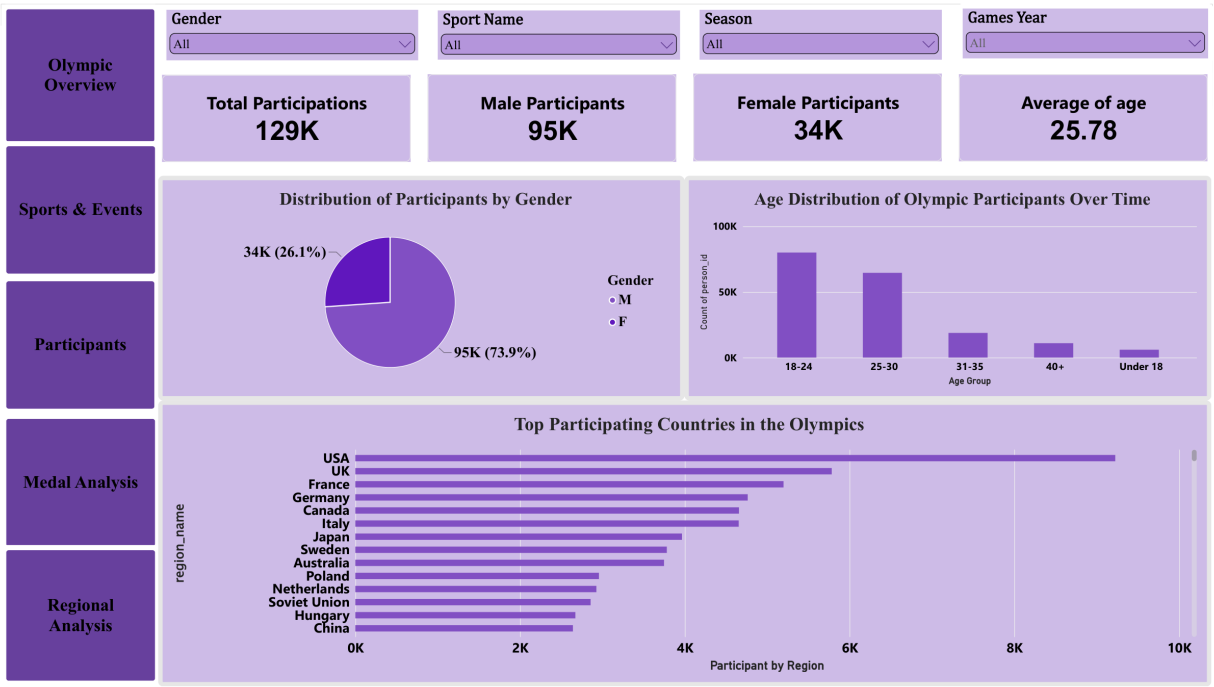
Source: Sports Analysis Dashboard BI PDF supplied for this project. Reproduced without redesign.

Dashboard source page 2 of 5



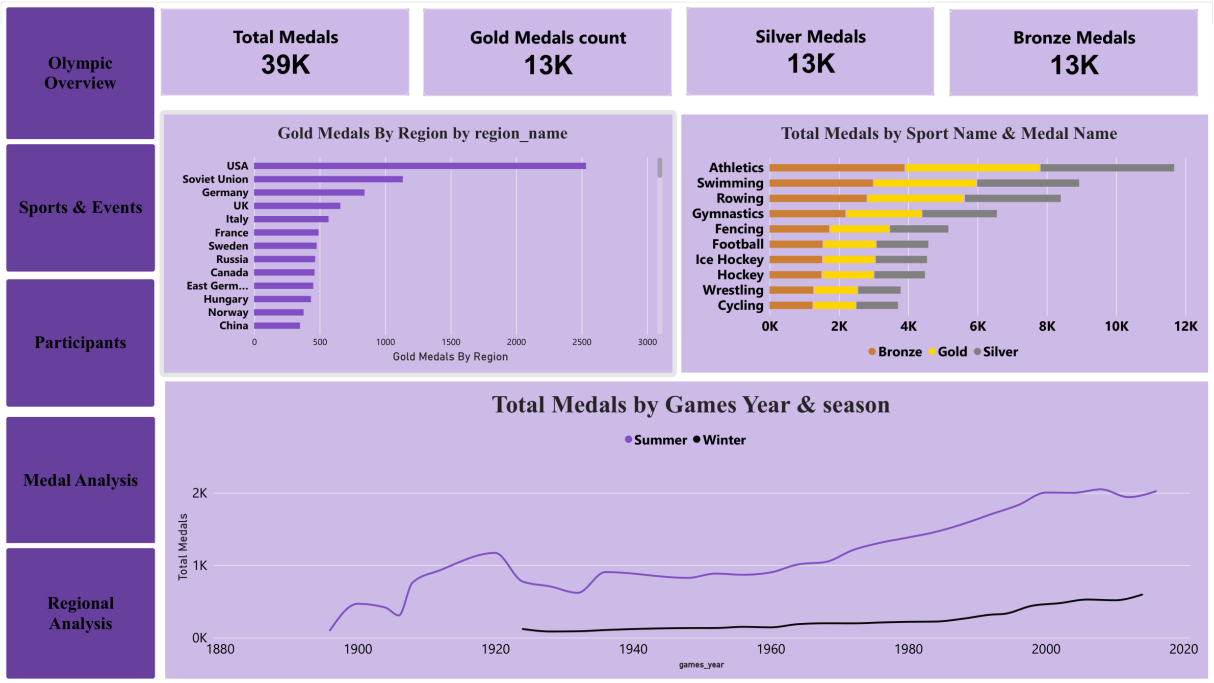
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Dashboard source page 3 of 5



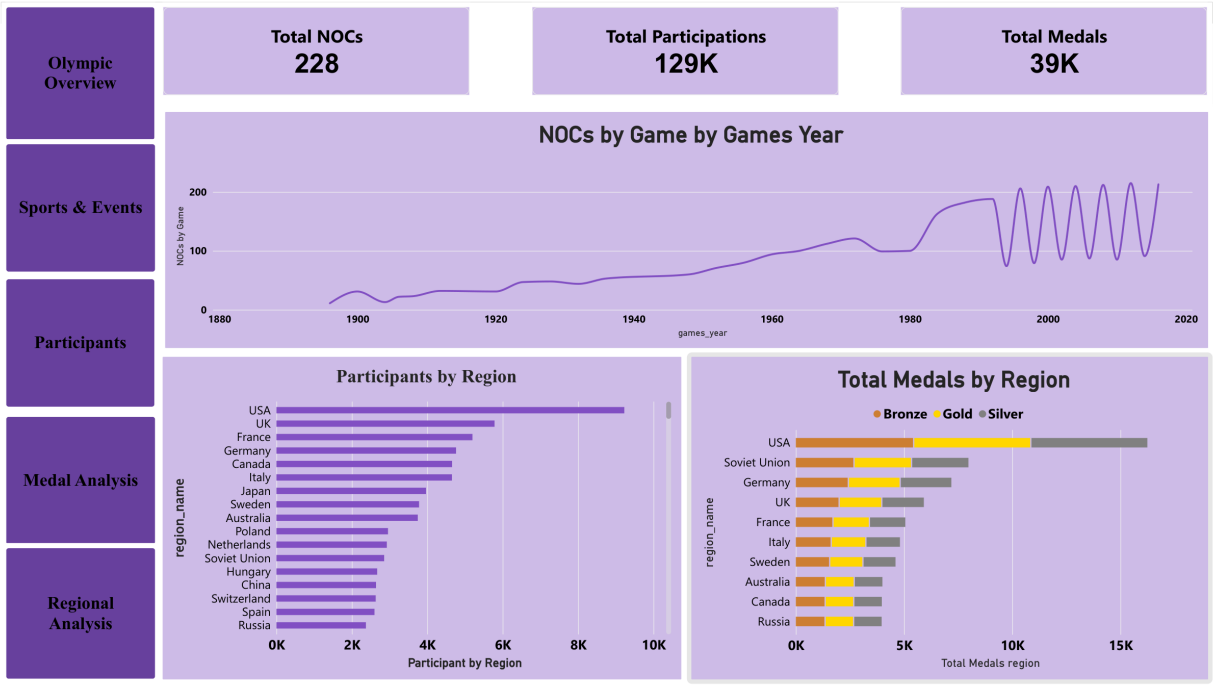
Source: Sports Analysis Dashboard BI PDF supplied for this project. Reproduced without redesign.

Dashboard source page 4 of 5



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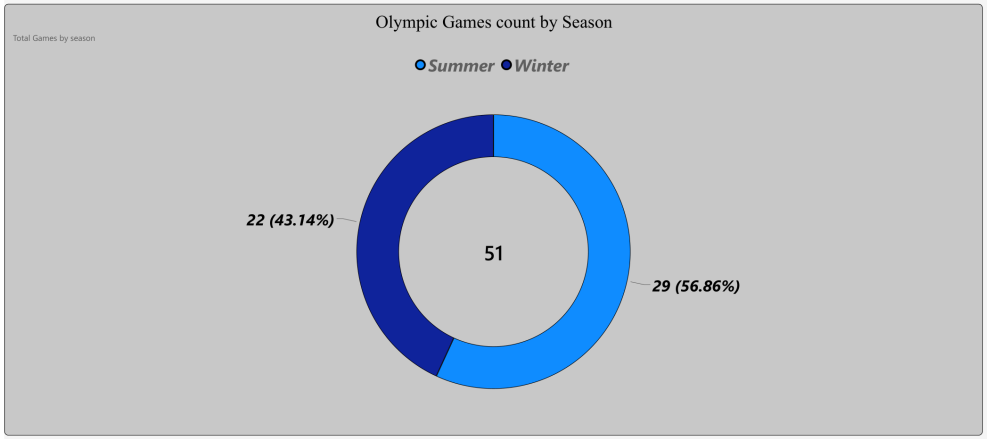
Dashboard source page 5 of 5



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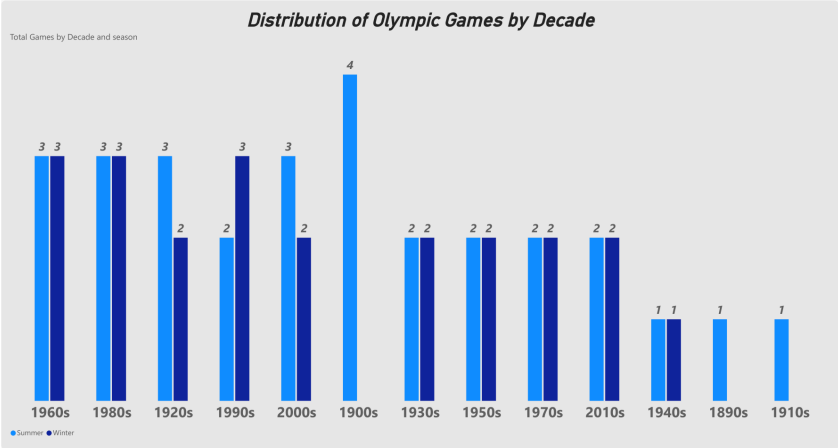
Solved question 1

1. How many Olympic Games have been held in each season (Summer vs. Winter)?



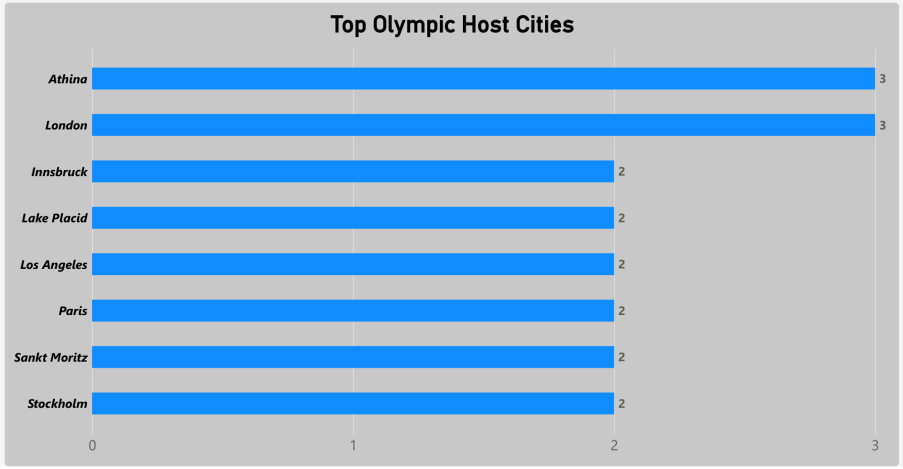
Solved question 2

2 What is the distribution of games across different decades?



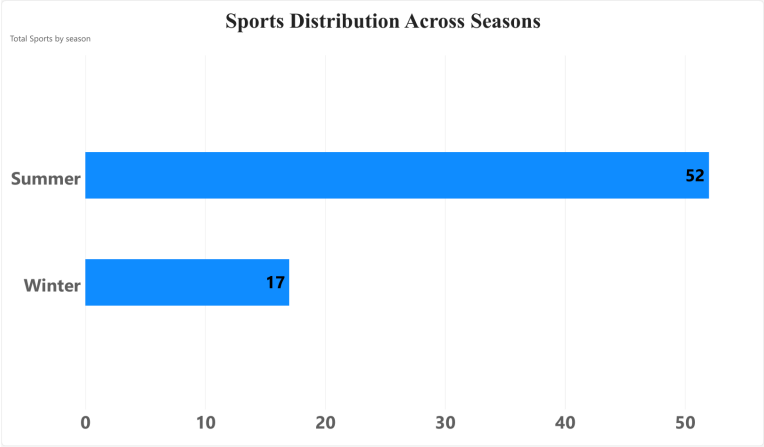
Solved question 3

3 Which cities have hosted the most Olympic Games?



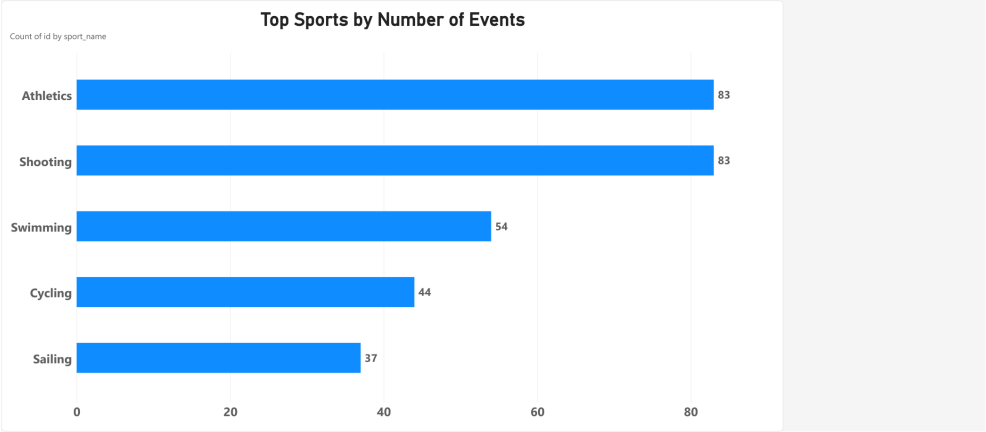
Solved question 4

4 What is the distribution of sports between Summer and Winter Olympics?



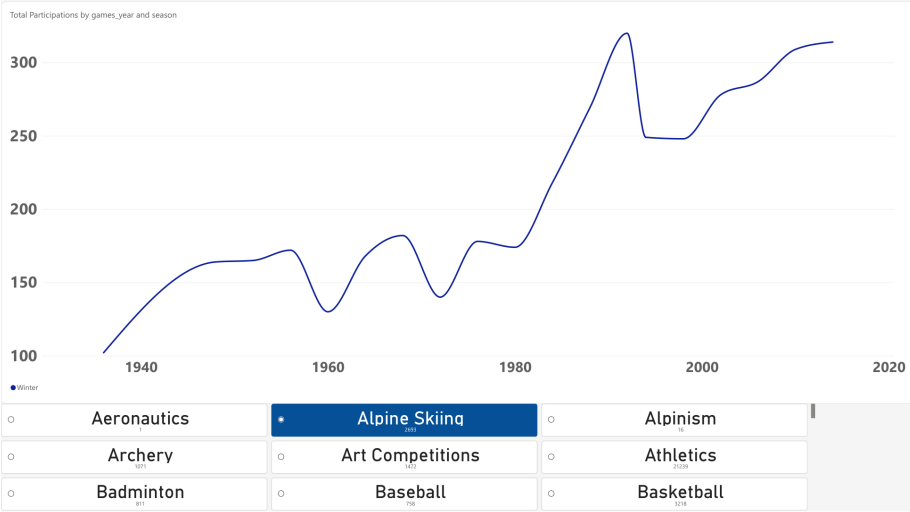
Solved question 5

5 Which sports have the highest number of events in the Olympics?



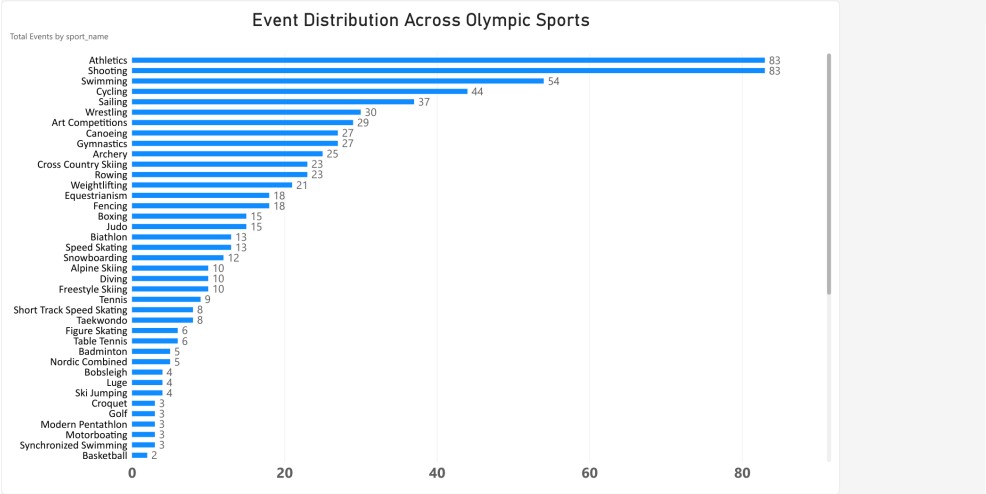
Solved question 6

6 How has the participation in each sport evolved over time?



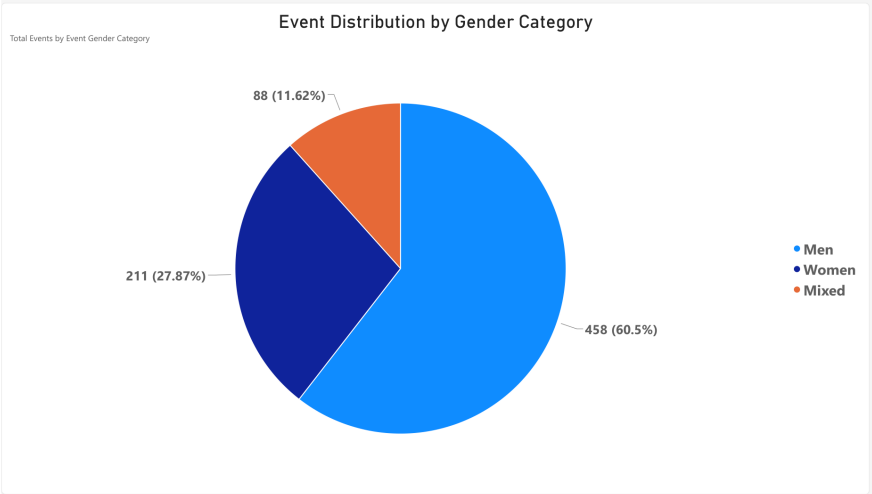
Solved question 7

7 How many events are there in each sport?



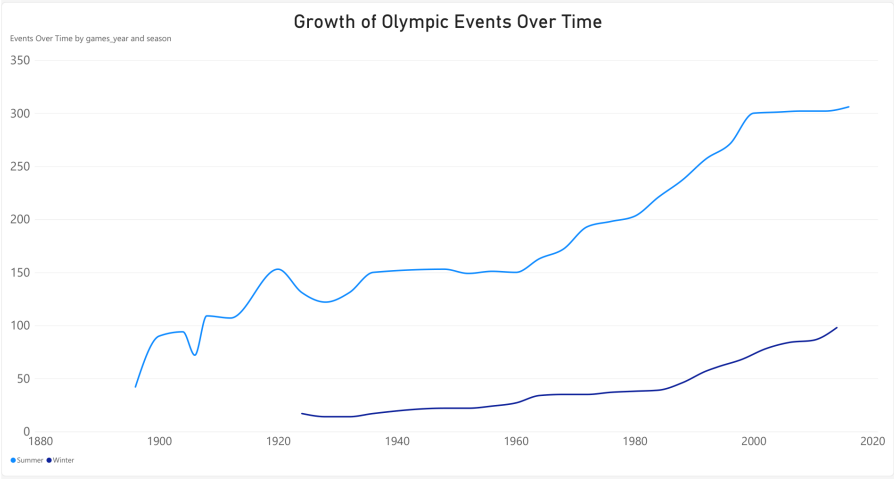
Solved question 8

8 What is the distribution of events by gender (Men, Women, Mixed)?



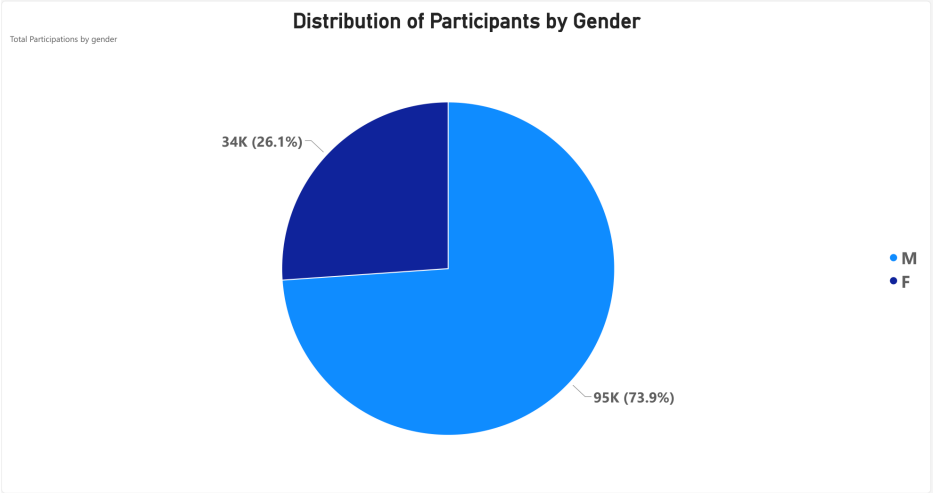
Solved question 9

9 How has the number of events changed over time?



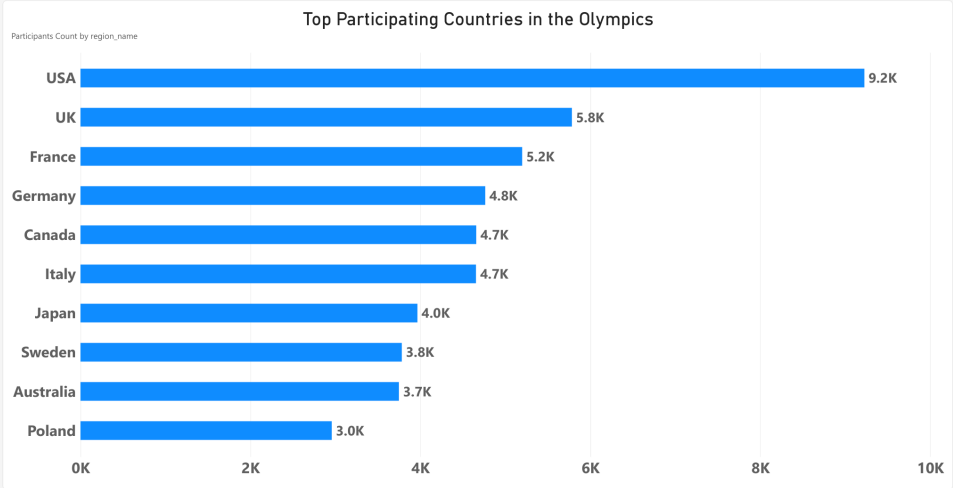
Solved question 10

10 What is the distribution of participants by gender?



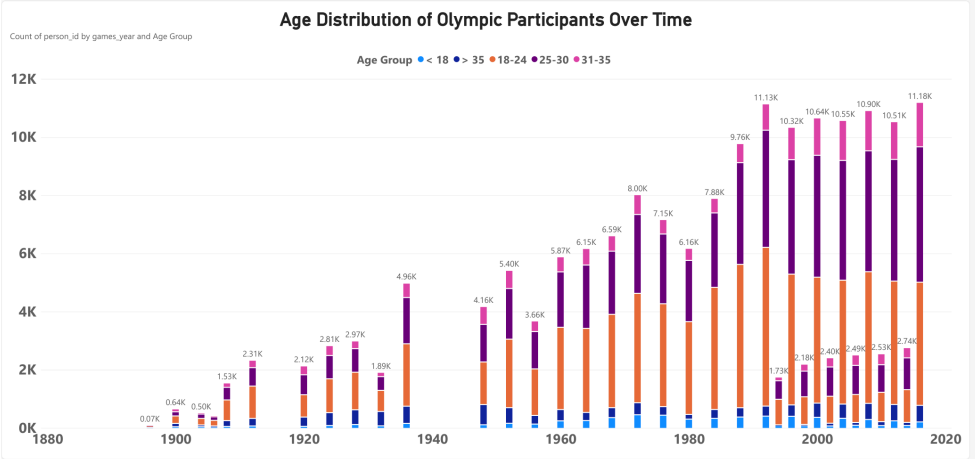
Solved question 11

11 Which countries have the highest number of participants in the Olympics?



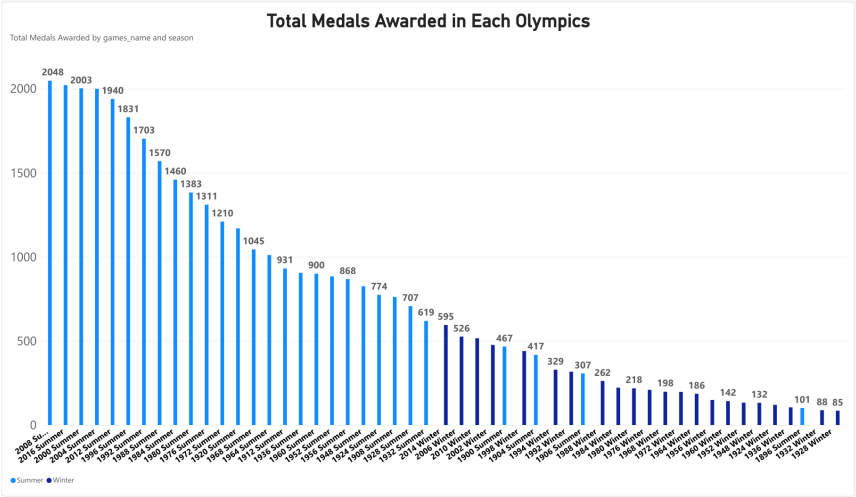
Solved question 12

12 How does the age distribution of participants vary across different games?



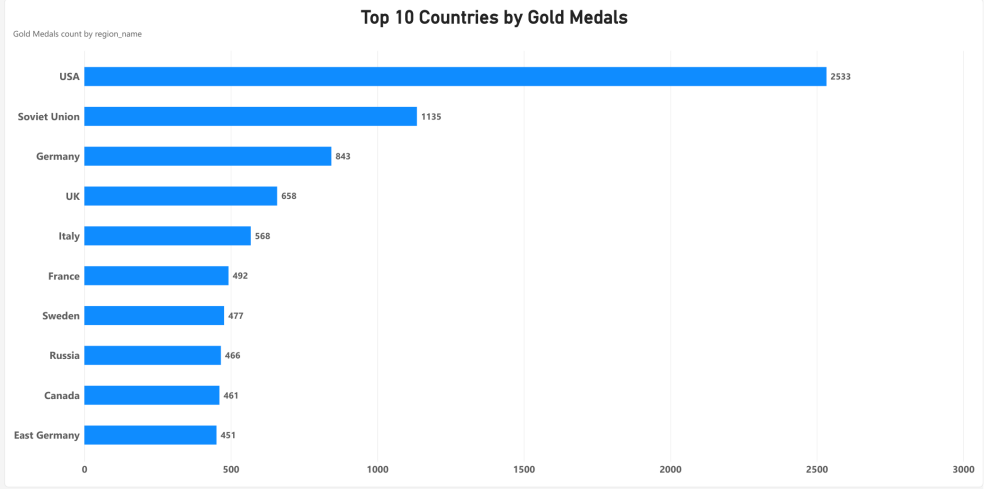
Solved question 13

13 How many medals have been awarded in each Olympics?



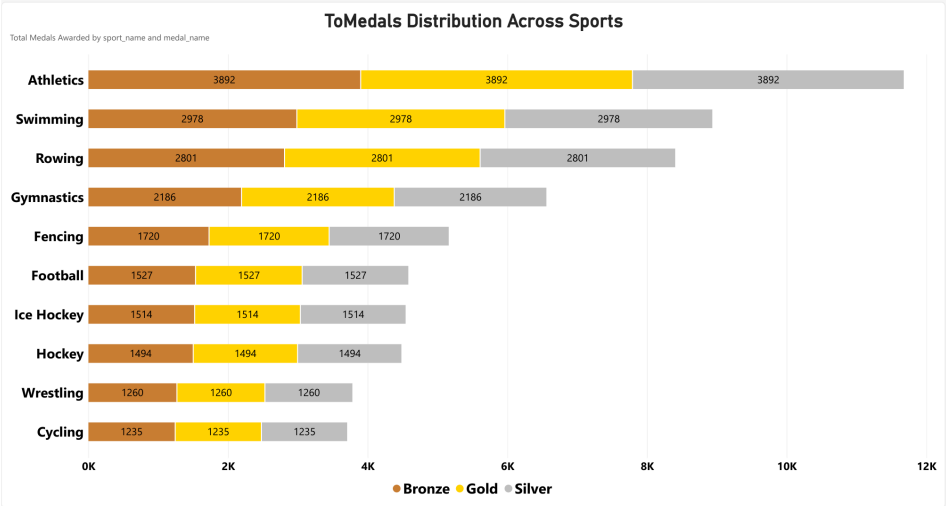
Solved question 14

14 Which countries have the highest number of gold medals?



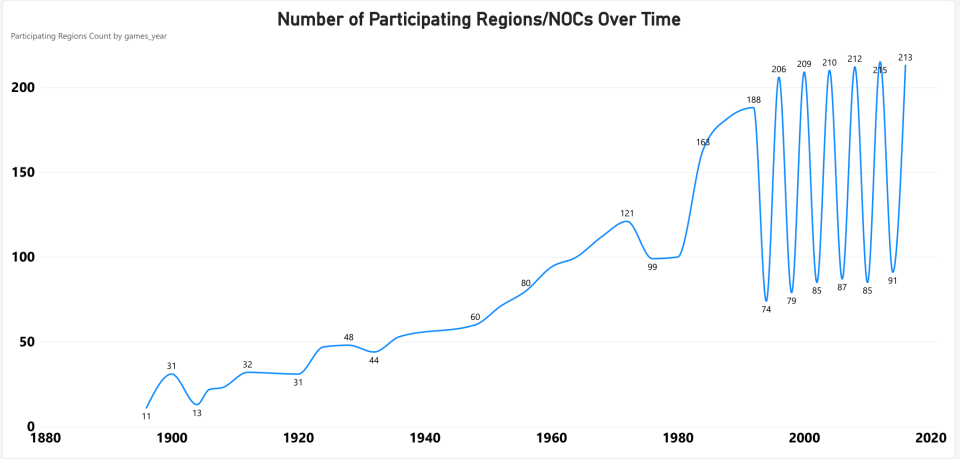
Solved question 15

15 How does the medal distribution vary across different sports?



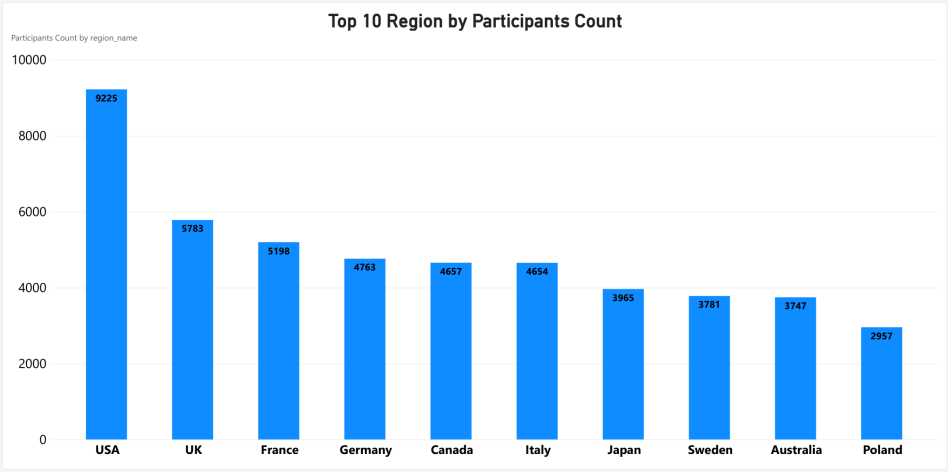
Solved question 16

16 How many regions or NOCs participate in each Olympic Games?



Solved question 17

17 Which regions have the highest number of participants in the Olympics?



Solved question 18

What is the distribution of medals among different regions?

